

Title: Linear Regression Model — House Price Predictor Internship Task 1 | Maincrafts Technology

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1. Introduction

The objective of this project was to build a machine learning model to predict house prices in California using Linear Regression. This project covers the complete ML workflow including data loading, exploratory data analysis, preprocessing, model training, evaluation, and visualization.

2. Dataset

The California Housing dataset was used, which is built into the scikit-learn library. It contains 20,640 rows and 9 features including:

- MedInc (Median Income)
- HouseAge
- AveRooms
- AveBedrms
- Population
- AveOccup
- Latitude & Longitude
- MedHouseVal (Target — House Price)

3. EDA Findings

- No missing values were found in the dataset
- The dataset contains 20,640 houses across California
- MedInc (income) has the highest correlation of 0.69 with house price, meaning higher income areas tend to have higher house prices
- Latitude and Longitude had a strong negative correlation of -0.92 with each other

4. Model Training

- Data was split 80% for training (16,512 rows) and 20% for testing (4,128 rows)
- A Linear Regression model was trained using scikit-learn

5. Model Results

Metric

Value

MAE

0.533

RMSE

0.746

R² Score

0.576

The model explains 57.6% of the variance in house prices. The average prediction error is \$53,300.

6. Improvement Ideas

- Apply feature scaling using StandardScaler to improve model performance
- Try Ridge or Lasso Regression to reduce overfitting
- Use more advanced models like Random Forest or XGBoost
- Engineer new features such as rooms per person

7. Conclusion

A Linear Regression model was successfully built and evaluated on the California Housing dataset. The model achieved an R² score of 0.576, which is a reasonable baseline. Future improvements can be made using advanced algorithms and feature engineering.

